

Problem Sheet #1

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Course: Measure Theory
Date: 2026-04-06

– Problem 1:

If A and B are sets, show that $A \subseteq B$ if and only if $A \cap B = A$.

— *Solution:*

We proceed by mutual implication.

Forward Direction ($\Rightarrow$): Assume $A \subseteq B$. We must show $A \cap B = A$. First, let $x \in A \cap B$. By the definition of intersection, $x \in A$ and $x \in B$. Thus, $x \in A$, which implies $A \cap B \subseteq A$. Second, let $x \in A$. Since $A \subseteq B$, it follows that $x \in B$. Because $x \in A$ and $x \in B$, we have $x \in A \cap B$. Thus, $A \subseteq A \cap B$. By mutual inclusion, $A \cap B = A$.

Reverse Direction ($\Leftarrow$): Assume $A \cap B = A$. We know by that $A \cap B \subseteq B$, using our assumption we conclude that $A \subseteq B$

Q.E.D.

– Problem 2:

If A is a set and $\mathcal{B} = \{B_i : i \in I\}$ is any family of sets, show that:

$$A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i) \text{ and } A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i)$$

— *Solution:*

First part.

Forward Direction ($\Rightarrow$): Let $x \in (A \cap \bigcup_{i \in I} B_i)$.

By definition of intersection we have $x \in A \wedge \exists j \in I$ such that $x \in B_j$, so we have that $x \in A \cap B_j$, if this element is in the intersection of a sole set of the family, is also true that is in the union of all sets in the family, therefore $x \in \bigcup_{i \in I} (A \cap B_i)$

Reverse Direction ($\Leftarrow$): Let $x \in \bigcup_{i \in I} (A \cap B_i)$

We know that $\exists j \in I$ such that $x \in B_j \cap A$, this is $x \in A \wedge x \in B_j$, because x is in any set of the family of sets, is also true that is in the union of the family, therefore $x \in \bigcup_{i \in I} B_i \wedge x \in A$, so we have that $x \in A \cap \bigcup_{i \in I} B_i$

Second part.

Forward Direction ($\Rightarrow$): Let $x \in A \cup \bigcap_{i \in I} B_i$

So $x \in A \vee \forall j \in I, x \in B_j$ this is $x \in A \cup B_j$, for this we know that $\forall j \in I, x \in A \cup B_j$, this is $\bigcap_{i \in I} (A \cup B_i)$

Reverse Direction ($\Leftarrow$): Let $x \in \bigcap_{i \in I} (A \cup B_i)$

so $\forall j \in I, x \in A \cup B_j$, so if $\forall j \in I, x \in A \vee x \in B_j$, by Law of excluded middle we divide this in two cases:

1. $x \notin A$

- If this is the case, then $x \in B_j, \forall j \in I$ must be true, but this is trivially $x \in A \cup \bigcap_{i \in I} B_i$

2. $x \in A$

- This is trivially $x \in A \vee \forall j \in I, x \in B_j$ therefore $x \in A \cup \bigcap_{i \in I} B_i$

Q.E.D.

– **Problem 3:**

Let B be any set. Show that $(A \cap B) \cup C = A \cap (B \cup C) \iff C \subseteq A$.

— **Solution:**

Forward Direction ($\Rightarrow$): Assume $(A \cap B) \cup C = A \cap (B \cup C)$.

Let $x \in C$, by the definition of the union it follows that $x \in (A \cap B) \cup C$, we know by assumption that this is equal to $x \in A \cap (B \cup C)$, so $x \in A \wedge x \in (B \cup C)$, for this expression hold true we strictly require $x \in A$, therefore $x \in C \Rightarrow x \in A$ thus $C \subseteq A$

Reverse Direction ($\Leftarrow$): Assume $C \subseteq A$.

- **Forward inclusion:** $(A \cap B) \cup C \subseteq A \cap (B \cup C)$

• Let $x \in (A \cap B) \cup C$, this is $x \in A \cap B \vee x \in C$.

- Case 1: $x \in A \cap B$, we have that $x \in A \wedge x \in B$, and for definition of union it holds that $x \in B \cup C$, therefore $x \in A \cap (B \cup C)$
- Case 2: $x \in C$, for our assumption we know that this implies $x \in A$, also we know for definition of union $x \in B \cup C$, so $x \in A \wedge x \in B \cup C$, therefore $x \in A \cap (B \cup C)$

- **Backward inclusion:** $(A \cap B) \cup C \supseteq A \cap (B \cup C)$

• Let $x \in A \cap (B \cup C)$, this is $x \in A \wedge x \in B \cup C$

- Case 1: $x \in B$, we know that $x \in A$, this means $x \in A \cap B$, and for the definition of union $x \in (A \cap B) \cup C$ holds true
- Case 2: $x \notin B$, therefore $x \in C$, so by definition of union $x \in (A \cap B) \cup C$

Q.E.D.

– **Problem 4:**

The symmetric difference of two sets A and B is the set D of all elements that belong to either A or B but not both, symbolically,

$$D \equiv A \Delta B = (A \cup B) - (A \cap B)$$

Represent D with a diagram and show that $D = (A - B) \cup (B - A)$.

Furthermore, show that

i) $A \subseteq A \Delta B \cup A$,

ii) $A \subseteq A \Delta B \cup B$,

iii) $B \subseteq A \Delta B \cup A$,

iv) $B \subseteq A \Delta B \cup B$.

— **Solution:**

Show that $D = (A - B) \cup (B - A)$:

Let $x \in D$, this is $x \in (A \cup B) - (A \cap B)$, so we have

$$x \in A \cup B \wedge x \notin A \cap B \implies x \in A \cup B \wedge x \in A^c \cup B^c$$

this is

$$(x \in A \wedge x \in A^c) \vee (x \in B \wedge x \in A^c) \bigwedge (x \in A \wedge x \in B^c) \vee (x \in B \wedge x \in B^c)$$

In the right side

$$(x \in A \wedge x \in A^c) \vee (x \in B \wedge x \in A^c) \iff \text{FALSE} \vee (x \in B \wedge x \in A^c) \iff x \in B \wedge x \in A^c \\ \iff x \in B \wedge x \notin A \iff x \in (B - A)$$

In the left side

$$(x \in A \wedge x \in B^c) \vee (x \in B \wedge x \in B^c) \iff (x \in A \wedge x \in B^c) \vee \text{FALSE} \iff x \in A \wedge x \in B^c \\ \iff x \in A \wedge x \notin B \iff x \in (A - B)$$

joining both sides we got

$$(A - B) \cup (B - A)$$

i) $A \subseteq A \Delta B \cup A$,

Let $x \in A$, for the definition of union $x \in A \Delta B \cup A$

ii) $A \subseteq A \Delta B \cup B$,

Let $x \in A$, thus $x \in A \cup B$ by definition of union. We have two cases.

Case 1: $x \in B$ therefore $x \in A \Delta B \cup B$, by the definition of union.

Case 2: $x \notin B$, we can't go directly as first case, instead we denote $x \in A \wedge x \notin B \iff x \in A - B$ therefore by the first demonstration we got that $x \in (A - B) \cup (B - A) \iff x \in A \Delta B \implies x \in A \Delta B \cup B$

iii) $B \subseteq A \Delta B \cup A$,

We proceed like in ii), let $x \in B$

iv) $B \subseteq A \Delta B \cup B$.

We proceed like in i) letting $x \in B$

Q.E.D.

– **Problem 5:**

Show that if $f : A \rightarrow B$ and $E, F \subseteq A$ then $f(E \cup F) = f(E) \cup f(F)$ and $f(E \cap F) \subseteq f(E) \cap f(F)$.

— **Solution:**

First part:

Forward Direction ($\Rightarrow$): Let $y \in f(E \cup F)$, by the definition of the function we have $\exists x \in E \cup F$ such that $f(x) = y$ thus $x \in E \vee x \in F$,

Case 1: $x \in E \Rightarrow f(x) \in f(E) \Leftrightarrow y \in f(E)$

Case 2: $x \in F \Rightarrow f(x) \in f(F) \Leftrightarrow y \in f(F)$

For this we have that $y \in f(E) \vee y \in f(F) \Rightarrow y \in f(E) \cup f(F)$ thus $f(E \cup F) \subseteq f(E) \cup f(F)$

Backward Direction ($\Leftarrow$): Let $y \in f(E) \cup f(F) \Leftrightarrow y \in f(E) \vee y \in f(F)$ thus:

Case 1:

Let $y \in f(E) \Rightarrow \exists x_E \in E$ such that $f(x_E) = y$ by definition of union $x_E \in E \cup F$ therefore $f(x_E) \in f(E \cup F) \Leftrightarrow y \in f(E \cup F)$

Case 2:

Let $y \in f(F) \exists x_F \in F$ such that $f(x_F) = y \Rightarrow x_F \in E \cup F$ therefore $f(x_F) \in f(E \cup F) \Leftrightarrow y \in f(E \cup F)$

so we know that $f(E) \cup f(F) \subseteq f(E \cup F)$

So we demonstrate that $f(E \cup F) = f(E) \cup f(F)$

Second part:

Let $y \in f(E \cap F)$ for definition of function $\exists x \in E \cap F$ such that $f(x) = y$ thus $x \in E \wedge x \in F$ we know that for $x \in E \Rightarrow f(x) \in f(E) \wedge x \in F \Rightarrow f(x) \in f(F)$ therefore $f(x) \in f(E) \cap f(F) \Leftrightarrow y \in f(E) \cap f(F)$ so we conclude that $f(E \cap F) \subseteq f(E) \cap f(F)$

Q.E.D.

– **Problem 6:**

Show that if $f : A \rightarrow B$ and G, H are subset of B , then $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ and $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$

— **Solution:**

First part: Let $a \in f^{-1}(G \cup H) \Rightarrow f(a) \in G \cup H$

Case 1: $f(a) \in G \Leftrightarrow a \in f^{-1}(G)$

Case 2: $f(a) \in H \Leftrightarrow a \in f^{-1}(H)$

Thus $a \in f^{-1}(G) \vee a \in f^{-1}(H) \Leftrightarrow a \in f^{-1}(G) \cup f^{-1}(H)$

$\therefore f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H)$

Let $a \in f^{-1}(G) \cup f^{-1}(H) \Leftrightarrow a \in f^{-1}(G) \vee f^{-1}(H) \Leftrightarrow f(a) \in G \vee f(a) \in H$

Case 1: $f(a) \in G \Leftrightarrow f(a) \in G \cup H$ by definition of union, thus $a \in f^{-1}(G \cup H)$

Case 2: $f(a) \in H \Leftrightarrow f(a) \in G \cup H$ by definition of union, thus $a \in f^{-1}(G \cup H)$

$$\therefore f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$$

$$\text{So we got: } f^{-1}(G) \cup f^{-1}(H) = f^{-1}(G \cup H)$$

Second part:

Let $a \in f^{-1}(G \cap H)$ for definition of preimage we got

$f(a) \in G \cap H \iff f(a) \in G \wedge f(a) \in H$ applying again definition of preimage we got

$$a \in f^{-1}(G) \wedge a \in f^{-1}(H) \iff a \in f^{-1}(G) \cap f^{-1}(H)$$

$$\therefore f^{-1}(G \cap H) \subseteq f^{-1}(G) \cap f^{-1}(H)$$

$$\text{Let } a \in f^{-1}(G) \cap f^{-1}(H) \iff a \in f^{-1}(G) \wedge a \in f^{-1}(H)$$

by definition of preimage $f(a) \in G \wedge f(a) \in H \iff f(a) \in G \cap H$ thus $a \in f^{-1}(G \cap H)$

$$\therefore f^{-1}(G \cap H) \supseteq f^{-1}(G) \cap f^{-1}(H)$$

$$\text{So } f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$$

Q.E.D

– **Problem 7:**

Show that if $f : A \rightarrow B$ is injective and $E \subseteq A$ [respectively, surjective and $H \subseteq B$], then $f^{-1}(f(E)) = E$ [respectively, $f(f^{-1}(H)) = H$].

— **Solution:**

First part:

Let $x \in E \implies f(x) \in f(E)$ and by definition of preimage $x \in f^{-1}(f(E))$ so we conclude that $E \subseteq f^{-1}(f(E))$

Let $x \in f^{-1}(f(E))$, this implies $f(x) \in f(E)$ thus $\exists y \in E$ such that $f(y) = f(x)$

recall the definition of injectiveness, $f(x_1) = f(x_2) \implies x_1 = x_2$ so if $f(y) = f(x) \implies x = y$, therefore $f(y) \in f(E) \implies y \in E \iff x \in E$

Second part:

Let $y \in f(f^{-1}(H))$ by definition of image $\exists x \in f^{-1}(H)$ such that $y = f(x)$, by definition of preimage it follows $f(x) \in H$, and this implies $y \in H$ therefore $f(f^{-1}(H)) \subseteq H$

Let $y \in H$, because $H \subseteq B$, we know that y is in the codomain, and using the definition of surjectivity we know that $\forall y \in H, \exists x \in A$ such that $y = f(x)$, thus we have $y \in H \iff f(x) \in H$ and applying definition of preimage $x \in f^{-1}(H)$, this implies by image definition that $f(x) \in f(f^{-1}(H)) \iff y \in f(f^{-1}(H))$

Q.E.D

– **Problem 8:**

Show that if f is a bijection from A onto B , then f^{-1} is a bijection of B onto A . Moreover, if g is a bijection of B onto C , then the composite function $g \circ f$ is a bijection of A onto C .

— **Solution:**